



Unsupervised Concept Discovery and Evaluation for Medical Vision-Language Models

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Presenter: Emmanuel.

Good morning everyone. Today we present our work on Unsupervised Concept Discovery and Evaluation for Medical Vision-Language Models, focusing on extracting interpretable features from chest X-ray embeddings.



Introduction

Why Are Medical VLMs Opaque?

- Medical **Vision-Language Models** (VLMs) encode images and text in a shared representation with strong clinical performance
- But their latent representations are **high-dimensional** and **polysemantic**
 - Each dimension mixes multiple clinical meanings
- This limits **interpretability** and **trust** in safety-critical applications
- **Concept-based explanations** aim to map representations onto recognizable anatomy and findings

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└ Introduction

Medical VLMs like BiomedCLIP encode images and text in a shared space with excellent performance. However, their latent representations are polysemantic: each dimension mixes multiple clinical meanings, making it hard to understand what the model has learned. This is a serious problem in safety-critical medical settings. Concept-based explanations try to solve this by mapping these opaque representations onto recognizable clinical concepts.

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Introduction

Our Approach

- Following **MedConcept**: extract SAE features **without diagnostic supervision**, name them, and evaluate against radiology reports
- **Adaptation** from the original pipeline:
 - 3D abdominal CT → **2D chest radiographs**
 - Backbone: **BiomedCLIP** (frozen ViT-B/16)
 - New vocabulary and activation selection
- **Objective**: Assess what the evidence supports about interpretability — *not* a diagnostic classifier

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Introduction

Following MedConcept, we use a Sparse Autoencoder to extract features without disease labels and evaluate them against real radiology reports. We adapt the pipeline from 3D abdominal CT to 2D chest X-rays, using BiomedCLIP as backbone with a new dedicated vocabulary. Important: our goal is not to build a classifier, but to assess what the evidence actually tells us about interpretability.

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Our Contributions

1. An implemented pipeline with **phrase-level semantic grounding**
2. Analysis of concept groups at **multiple clustering resolutions**
3. An independent **multi-evaluator audit** of the LLM judge

→ *Detailed results in the Experiments section*

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Introduction

Our three contributions are: a pipeline with phrase-level grounding, clustering analysis at multiple resolutions, and an independent audit of the LLM judge. The detailed results are coming up in the experiments section.

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Our Contributions

1. An implemented pipeline with **phrase-level semantic grounding**
2. Analysis of concept groups at **multiple clustering resolutions**
3. An independent **multi-evaluator audit** of the LLM judge

→ Detailed results in the Experiments section



Related Work

From Predefined Concepts to Unsupervised Discovery

- **Supervised concept methods** (TCAV, Concept Bottleneck Models): semantic interfaces, but require **predefined annotations**
- **Sparse dictionary learning**: decomposes polysemantic activations into monosemantic features *without* predefined concepts
- Recent approaches go further:
 - First *discover* features, then *name* them — decoupling representation from labeling
 - Decompose VLM embeddings directly into sparse combinations of existing textual concepts
- In the **medical domain**: recent works dissect neuron activations in diagnostic models to extract concept-level explanations from opaque networks

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Related Work

Traditional methods like TCAV and Concept Bottleneck Models need predefined annotations. Sparse dictionary learning removes this by extracting monosemantic features unsupervisedly. Recent approaches decouple discovery from naming, or decompose VLM embeddings directly into textual concepts. In the medical domain, several works extract interpretable concepts from opaque diagnostic networks, which motivates our approach.

Related Work
From Predefined Concepts to Unsupervised Discovery

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Research Gaps

Four Gaps Guide Our Analysis

1. Reconstruction $\neq$ Correct Naming

Vocabulary coverage and image-text geometry constrain naming

2. Dependence on External Resources

Ontology, generative phrasing, LLM judges — errors can be mistaken for representation properties

3. Reports as Selective Evidence

Omission $\neq$ absence — judges can confuse the two

4. Independent Labels Omit Relationships

Embedding similarity $\neq$ anatomical hierarchy

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4. **Independent Labels Omit Relationships**
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Four gaps guide our analysis. First: good reconstruction does not mean correct naming. The vocabulary and the embedding geometry constrain what labels are even possible. Second: the pipeline depends on external resources at every stage. Errors in these tools can look like properties of the representation itself. Third: reports are partial. Radiologists omit normal anatomy, and the judge can confuse omission with contradiction. Fourth: assigning labels independently ignores relationships between concepts.



Research Gaps

Our Response

- **Hierarchical clustering** at multiple resolutions to examine concept groups
- **Human + multi-model audit** on 25 stratified pairs to assess **evaluator reliability**
 - 3 human annotators + 3 commercial LLMs
- **Label permutation** to mitigate position bias in the LLM judge

→ *Goal: assess interpretability in a scalable and reproducible way*

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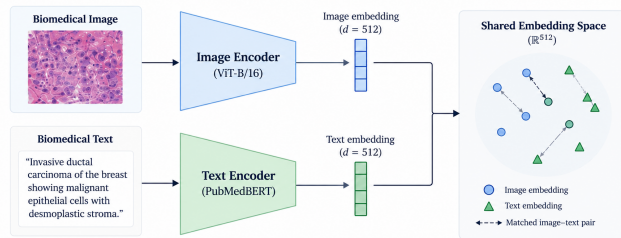
To address these gaps: we use hierarchical clustering to examine concept groups, a blinded audit with three annotators and three commercial LLMs to measure evaluator reliability, and label permutation to mitigate judge bias. The broader goal is assessing interpretability in a way that is scalable and reproducible.



Methodology

Datasets and Feature Extraction

- **NIH Chest X-ray 10k:** 7,500 SAE training / 2,500 validation images; no text or labels.
- **Open-I:** Evaluation pool of 3,666 frontal image-report pairs; Findings + Impression.
- **Features:** Frozen BiomedCLIP ViT-B/16 image encoder; **512-dimensional embeddings**.



BiomedCLIP overview

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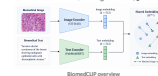
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Methodology

Concept Dictionary Construction

- **Concept Selection:** 100 parent concepts from cleaned and filtered Open-I Problems metadata, covering anatomy and pathological findings.
- **Expansion:** UMLS provides Concept Unique Identifiers (CUIs) and synonyms from SNOMED CT and MeSH for matched concepts.
- **Phrase Generation:** Gemini 3.1 Flash-Lite generates 5 radiology-style descriptions per concept.
- **Encoding:** BiomedCLIP's PubMedBERT text encoder maps all 998 dictionary entries to L2-normalized, 512-dimensional embeddings in the same space as the images.

$$T \in \mathbb{R}^{998 \times 512}$$

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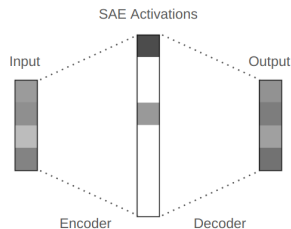
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Sparse Autoencoder Architecture and Training

- **Architecture:** 512-dimensional input, 1,024 hidden units (overcomplete factor of 2), ReLU activations z .
- **Objective:** Reconstruction error plus an L_1 penalty on activations; λ controls the sparsity penalty.

$$\mathcal{L} = \frac{1}{512B} \sum_{i=1}^B \|x_i - \hat{x}_i\|_2^2 + \frac{\lambda}{B} \sum_{i=1}^B \|z_i\|_1$$

- **Optimization:** Adam, batch size $B = 256$. Validation loss guides early stopping and model selection.



Illustrative SAE architecture

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Methodology

Concept Assignment and Inference

After SAE training, each decoder column represents a learned feature direction in the image embedding space.

- **Naming:** Compare normalized decoder columns $\tilde{W}_d$ with dictionary embeddings T . Assign the entry with the highest cosine similarity to each feature.

$$S = T\tilde{W}_d \in \mathbb{R}^{998 \times 1024}$$

- **Inference:** Rank positive SAE activations for each Open-I image. Select features from largest to smallest until cumulative activation $> 90\%$ of the total positive activation.
- **Output:** Collect the selected features' labels and remove duplicate phrases to obtain the predicted concept set.

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Semantic Evaluation and Validation

- **Evaluator:** Frozen MedGemma 1.5 4B receives a concept phrase and its reference report, without the image.
- **Verdicts:**
 - ✓ **Aligned:** Report explicitly supports the concept.
 - ✗ **Unaligned:** Report explicitly negates or contradicts the concept.
 - ? **Uncertain:** Concept is unmentioned, ambiguous, or tentative.
- **Bias Mitigation:** Rotate class–letter mappings across 3 passes; select the verdict with the highest average probability.
- **Clustering:** Group parent concepts with hierarchical clustering and compare alignment across groups using existing verdicts.
- **Validation:** 25 pairs independently labelled by 3 authors, blinded to MedGemma’s verdicts. Compare MedGemma and 3 commercial models against majority vote.

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Experiments

Sparse Reconstruction and Grounding

- **Reconstruction Fidelity:** Mean Squared Error (MSE) of 3.32×10^{-5} and mean cosine similarity of 0.9916 on the validation set[cite: 3].
- **Sparsity:** Average of 44.0 active units out of 1024 ($\sim 4.3\%$ L_0 sparsity) per image[cite: 3].
- **Takeaway:** The SAE achieves near-lossless visual reconstruction under a strict sparse activation regime[cite: 3]. However, fidelity does not guarantee semantic monosemanticity[cite: 3].

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Experiments

Moving to our results, the SAE training was highly successful in terms of reconstruction[cite: 3]. We achieved a very low Mean Squared Error and a cosine similarity above 0.99 on the validation set[cite: 3]. It did this while maintaining high sparsity, using an average of only 44 active units out of 1024 per image[cite: 3]. However, as we will see, this mathematical fidelity does not automatically mean the features are perfectly interpretable clinically[cite: 3].

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Experiments

Report Alignment and Cohort Analysis

- **Cohort Performance** ($N = 500$):
 - **Uncertain:** 70.30%[\[cite: 3\]](#)
 - **Unaligned:** 22.49%[\[cite: 3\]](#)
 - **Aligned:** 7.21%[\[cite: 3\]](#)
- **Omission Bias:** The dominance of Uncertain verdicts is due to radiologists omitting normal anatomy and unmentioned concepts[\[cite: 3\]](#).
- **Unaligned Rate:** Captures both false-positive concept predictions and explicit clinical negations[\[cite: 3\]](#).

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When we tested the extracted concepts against actual reports on a 500-image cohort, the alignment was surprisingly low, averaging about 7.2 percent[\[cite: 3\]](#). Over 70 percent of concepts were marked as Uncertain[\[cite: 3\]](#). This is heavily driven by reporting omission bias: radiologists document acute pathology but skip normal anatomy, which the SAE still extracts[\[cite: 3\]](#). The 22.5 percent Unaligned rate includes both model false-positives and explicit negations in the text[\[cite: 3\]](#).



Experiments

Hierarchical Concept Clustering

- **Approach:** Agglomerative hierarchical clustering on the 100 parent concept embeddings using cosine distance[cite: 3].
- **Localized Alignment:**
 - Focal, specific clusters score highly: *Costophrenic Angle / Sulcus* achieves **49.23%** Aligned[cite: 3].
 - Broad groups score poorly: *Mixed Cardiopulmonary* yields only 8.62% Aligned[cite: 3].
- **Conclusion:** Clinical agreement is highly localized and unevenly distributed across subdomains[cite: 3].

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To see if alignment varied by topic, we clustered the 100 parent concepts[cite: 3]. We found that clinical agreement is highly localized[cite: 3]. For example, focal anatomical findings like the "Costophrenic Angle" achieved over 49 percent alignment[cite: 3]. Conversely, broad or mixed clusters performed very poorly[cite: 3]. This shows that the SAE learns certain specific features much better than general ones[cite: 3].



Experiments

Evaluator Audit and Qualitative Case Studies

- **Human Consensus:** 3 human experts audited 25 concept-report pairs[cite: 3].
- **Model Agreement vs. Humans:**
 - **MedGemma 1.5 4B:** 56.0% agreement[cite: 3].
 - **Commercial LLMs** (GPT-5.6, Claude 5, Gemini 3.1 Pro): 68.0% — 76.0% agreement[cite: 3].
- **Failure Modes:** MedGemma suffered from negation parsing errors (e.g., misclassifying "no pleural effusion" as an Aligned "large effusion")[cite: 3].

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Finally, we audited our judge model[cite: 3]. Three experts manually evaluated 25 concept-report pairs[cite: 3]. We discovered that MedGemma agreed with human consensus only 56 percent of the time, while commercial models like Gemini 3.1 Pro reached up to 76 percent[cite: 3]. A major failure mode for MedGemma was negation: it would read "no pleural effusion" and incorrectly mark the concept "large effusion" as Aligned[cite: 3]. This proves that automated judges cannot yet replace expert review[cite: 3].

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Thank you for listening!
Any questions?